

Firm-level price wedges: how good is the mapping, and can it be improved?

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What is being measured

A firm's price wedge is not estimated directly. Wedges are estimated for decile portfolios, and a mapping carries those onto individual firms through their characteristics. The question is how much that mapping can be trusted, and whether it can be improved.

Four criteria, because they answer different questions.

Leaving one characteristic out. Fit on the portfolios of all but one characteristic, predict that one's ten deciles, rotate. Can the mapping price an anomaly it has not seen?

Leaving one family out. The same, holding out a whole Freyberger–Neuhierl–Weber category. Harder and more honest: value characteristics predict each other, so holding out book-to-market while assets-to-market and earnings-to-price remain is not a real test.

Holding out the extreme deciles. Fit on deciles two to nine, predict one and ten. This tests extrapolation, which is what the mapping must do at the firm level: 94% of firm-months sit outside the range the portfolios span on at least one characteristic.

Subsequent returns. Do firms the mapping calls overpriced go on to underperform? This is the only criterion evaluated on firms rather than portfolios, so it is the only one that can penalise bad extrapolation.

Everything is scored on magnitudes, not only ordering: the realised wedge is regressed on the fitted one and the intercept and slope reported. A perfect mapping has intercept zero and slope one. A mapping that ranks firms correctly but compresses the spread shows a slope away from one, and would be no good for a website that reports a wedge rather than a score.

Which portfolios count. Only characteristics whose construction is verified against the replication package enter, as regressors *and* as portfolios. A portfolio we cannot reproduce carries a wedge we cannot trust, so including it adds noise to the left-hand side as well as the right. With 1 characteristic still unverified (aPM), the rule costs nothing: the leave-one-family-out fit of the direct mapping is 0.63 with every characteristic in and 0.62 without it. When six were unverified it raised that fit from 0.56 to 0.67.

Results

Leave one characteristic out

Mapping	R^2	Correlation	Intercept	Slope	RMSE (pp)
3 PCs (paper)	0.587	0.766	-0.19	0.959	6.23
10 PCs	0.747	0.864	-0.12	0.965	4.88
direct	0.766	0.875	-0.11	0.959	4.70
direct, precision-wtd	0.774	0.880	-0.35	0.944	4.63
direct + squares	0.370	0.608	-1.39	0.626	8.48
squares, precision-wtd	0.449	0.670	-1.31	0.640	8.09

Leave one family out

Mapping	R^2	Correlation	Intercept	Slope	RMSE (pp)
3 PCs (paper)	0.389	0.624	-1.68	0.665	8.15
10 PCs	0.467	0.683	-1.70	0.673	7.76
direct	0.619	0.787	-1.01	0.843	6.14
direct, precision-wtd	0.609	0.781	-1.30	0.805	6.33
direct + squares	0.038	0.195	-4.89	0.047	39.54
squares, precision-wtd	0.049	0.222	-4.88	0.056	37.44

Extremes held out

Mapping	R^2	Correlation	Intercept	Slope	RMSE (pp)
3 PCs (paper)	0.737	0.858	+1.02	0.963	7.23
10 PCs	0.805	0.897	+0.04	0.863	6.46
direct	0.804	0.897	-1.35	0.853	6.56
direct, precision-wtd	0.808	0.899	-1.37	0.856	6.50
direct + squares	0.571	0.756	+0.11	0.798	9.51
squares, precision-wtd	0.626	0.791	-1.52	0.699	9.72

Do the wedges predict returns?

Firms are sorted into deciles on their fitted wedge each month and tracked forward. Decile 1 is the most negative wedge (most underpriced), decile 10 the most positive, so if wedges are mispricing that resolves, decile 1 should outperform and the spread should be positive.

Mapping	Horizon	Weighting	Decile 1	Decile 10	Spread (pp)
3 PCs (paper)	12m	cap	18.4	10.7	7.7
3 PCs (paper)	36m	cap	60.3	34.8	25.5

Mapping	Horizon	Weighting	Decile 1	Decile 10	Spread (pp)
3 PCs (paper)	60m	cap	118.3	69.4	48.9
3 PCs (paper)	12m	equal	26.3	11.5	14.8
3 PCs (paper)	36m	equal	80.4	35.1	45.3
3 PCs (paper)	60m	equal	143.2	67.0	76.2
10 PCs	12m	cap	21.3	10.5	10.7
10 PCs	36m	cap	71.1	33.4	37.7
10 PCs	60m	cap	134.3	65.6	68.8
10 PCs	12m	equal	27.2	10.5	16.7
10 PCs	36m	equal	84.1	31.6	52.6
10 PCs	60m	equal	150.0	62.3	87.7
direct	12m	cap	18.9	10.1	8.8
direct	36m	cap	64.6	31.9	32.7
direct	60m	cap	125.1	62.4	62.8
direct	12m	equal	25.5	10.0	15.5
direct	36m	equal	80.4	30.2	50.2
direct	60m	equal	145.9	58.9	87.0

The wedges predict returns strongly and with the right sign. Over five years the direct mapping separates the extreme deciles by 63 percentage points capitalisation-weighted and 87 equal-weighted, and its deciles are monotone except capitalisation-weighted at 36 months. Three principal components produce a smaller spread (49 and 76 points) and are not monotone capitalisation-weighted at 12, 36 and 60 months.

Can the mapping be improved?

Precision weighting: no, and the reason is informative

If some portfolio wedges were estimated far more precisely than others, the regression should weight them accordingly. They are not. Reproducing the package's own standard error — a Newey–West variance on the cohort-level ratios carried through the logarithm by the delta method — and separately a moving-block bootstrap with the package's 180-month blocks:

	Median	10th pct	90th pct	Max
Standard error (pp)	29.0	27.1	32.2	40.6

The ratio of the 90th to the 10th percentile is 1.2. Precision is close to uniform across the 560 portfolios, so weighting by it has almost nothing to exploit — and in the event it is very slightly worse on all three criteria, presumably because the weights add estimation noise of their own.

What the exercise does reveal is the *level*: a standard error around 29 percentage points on wedges whose cross-sectional standard deviation is about 10. Individual portfolio wedges are imprecise, because 463 formation cohorts that overlap in 179 of their 180 months are worth only two or three independent observations. That is a caveat on the whole exercise rather than on any mapping.

Non-linear terms: no

Adding squared ranks roughly halves out-of-sample fit and pushes the slope to 0.6 or below. The design already carries the non-linearity that matters: using all ten deciles rather than the extremes lets the wedge be non-linear in each sorting characteristic, and squaring 56 standardised ranks mostly adds collinear regressors for 560 observations to overfit.

Shrinkage: yes, and it is the substance

Unpenalised, the direct regression fits best in sample and generalises worst. Penalised at the level chosen inside each fold, it is the best mapping on every criterion at once and is close to unbiased in magnitude. The penalty is not a technicality: without it the fitted wedges of unfamiliar portfolios are more than twice too dispersed. On the leave-one-family-out test the realised wedge moves 0.42 for each point of fitted wedge with no penalty, against 0.89 with the penalty of 100 (run of 12 September 2026, `group_cv.held.out` on the verified portfolios).

Number of components

Ten principal components come close to the direct regression and beat three decisively on every criterion. If a dimension-reduced mapping is wanted for interpretability, ten is the defensible choice; three is not.

Is the cross-section of wedges stable over time, and which sample should the site use?

Every portfolio wedge is $-\log$ of the mean, over formation cohorts, of the discounted fifteen-year cash flow per dollar of price. The full available sample has 559 cohorts, formed 196407 to 201101, with cash flows to 202512. Cohorts overlap in 179 of their 180 months, so the sample holds about three independent fifteen-year windows, and any subsample of it one or two. The question is whether the cross-section of wedges is the same object in different parts of the sample, and if not, which part the site should rest on.

`stability.py` re-estimates all 570 decile wedges on subsets of the cohorts: the paper's own window, halves, thirds, and every window that starts in a later year. On each subset the market price of risk is re-solved so that the market wedge is zero there, as it is on the full sample; without that, a subsample's wedges shift as a block with the market's realised return. All comparisons below use the 560 portfolios of the verified characteristics.

Each subsample against the full sample

Cohorts formed	N	λ	sd	noise	corr	rank	slope	RMSE	LS corr	flips
full 1964-2011	559	3.53	9.3	3.4	1.00	1.00	1.00	0.0	1.00	0
paper 1964-2002	462	3.97	10.0	3.4	0.99	0.99	1.06	1.4	0.99	1
first half 1964-1987	282	4.52	10.7	3.5	0.91	0.90	1.04	4.6	0.94	5
second half 1988-2011	277	2.86	10.3	3.4	0.76	0.76	0.83	6.9	0.78	8
third 1964-1979	186	5.58	12.1	3.0	0.83	0.81	1.08	7.3	0.85	8
third 1980-1995	192	2.49	5.6	3.2	0.17	0.18	0.10	10.9	-0.02	29
third 1996-2011	181	2.85	15.0	4.1	0.76	0.78	1.23	10.2	0.80	5

Cohorts formed	N	λ	sd	noise	corr	rank	slope	RMSE	LS corr	flips
from 1975	433	2.62	8.2	3.4	0.81	0.79	0.71	5.5	0.87	4
from 1980	373	2.64	8.6	3.1	0.76	0.75	0.70	6.3	0.80	7
from 1985	313	2.78	9.3	3.4	0.75	0.74	0.74	6.7	0.78	9
from 1990	253	2.90	11.3	3.2	0.77	0.77	0.93	7.4	0.79	5
from 1995	193	2.92	14.3	3.8	0.76	0.78	1.17	9.6	0.79	6
from 2000	133	2.16	13.9	4.4	0.52	0.53	0.77	12.2	0.49	11

sd: cross-sectional standard deviation of the subsample’s wedges, pp. *noise*: median idiosyncratic standard error of a wedge in that subsample, pp, from a moving-block bootstrap over its cohorts with each draw demeaned across portfolios so the component common to all portfolios drops out (block length 120 cohorts, or a third of the subsample when shorter; 300 draws). *corr*, *rank*, *slope*, *RMSE*: the subsample’s wedges against the full sample’s, slope of subsample on full. *LS corr* and *flips*: the correlation of the 56 long-short spreads with the full sample’s, and how many change sign.

Two things stand out. Adding the 2003–2011 cohorts to the paper’s window changes almost nothing: correlation 0.99, one sign flip, RMSE 1.4 pp. Dropping the early cohorts is a different matter. Every window that starts in 1980 or later correlates 0.5 to 0.8 with the full cross-section, and the 1980–1995 third is nearly flat (sd 5.6 pp against 9.3) and unrelated to the rest.

The two halves against each other

The halves are independent cohorts, so this is the cleanest test. Their wedges correlate 0.45 (rank 0.46), with a slope of second on first of 0.43. Given each half’s idiosyncratic noise, two halves of a *constant* cross-section would correlate about 0.68. The change across portfolios has a standard deviation of 11.0 pp, of which noise accounts for 5.3 pp and about 8.6 pp is left, on a cross-sectional spread of 9.3 pp. 27% of portfolios move by more than two idiosyncratic standard errors. So the cross-section moves between halves by roughly its own size. That is the honest reading, with one caveat in the other direction: a block bootstrap with blocks shorter than the 180-month overlap understates sampling noise, so some of the 8.6 pp may still be sampling variation. The gap between 0.45 and 0.68 is too large for all of it to be. Judged against the *total* Newey–West standard errors, which include the component common to all portfolios, no portfolio’s change is significant (median total noise 51 pp); that statistic answers a different question, whether a single wedge is known, and the answer to that was already no.

Where the change sits

Long-short spread of each characteristic (long leg less short leg, pp) by subsample, ordered by how much it moves between the halves. 77% of characteristics keep the sign of their spread across halves; the spreads correlate 0.54 across halves.

	full	paper	1964–87	1988–2011	1964–79	1980–95	1996–2011
D2P	-16	-23	-39	40	-43	-4	61
A2ME	-20	-27	-42	22	-47	-12	40
E2P	-26	-30	-51	12	-59	-14	18
ROC	-43	-49	-61	-14	-65	2	-27
C2D	10	13	19	-15	18	7	-25
SAT	-20	-15	-3	-34	11	0	-61
ROE	26	29	38	7	37	-15	23

	full	paper	1964–87	1988–2011	1964–79	1980–95	1996–2011
CAT	-14	-8	3	-28	17	0	-53
RNA	14	16	27	-4	28	4	-3
ATO	4	0	-13	17	-26	4	22
S2P	-42	-46	-49	-20	-47	3	-43
PROF	13	17	27	1	36	-23	12
OL	-21	-16	-6	-32	8	2	-61
AOA	-22	-24	-31	-7	-33	-5	-14
sdDVOL	-27	-26	-15	-38	-7	-13	-49
C2A	26	30	32	10	34	-13	26
NOA	-2	0	5	-17	9	-27	-9
ROIC	27	29	35	13	34	-7	25
R_12_2	23	24	13	33	15	9	48
ROA	23	27	32	12	33	-10	25
sdTURN	5	1	-2	16	-14	6	17
EPS	7	4	-4	13	-17	-14	23
R_12_7	23	24	14	31	16	10	44
TAN	17	17	16	-0	12	-6	9
R_6_2	14	15	7	22	8	3	33
TNOVR	-7	-11	-13	2	-27	5	-5
DTO	-6	-8	-12	1	-18	4	-5
AT	22	20	22	9	9	-6	13
aBEME	-52	-55	-57	-44	-63	7	-68
MAXRET	-3	-6	-1	-12	-12	-6	-21
PCM	26	27	28	17	23	-15	34
dSOUT	-38	-39	-34	-44	-36	-30	-54
IVC	-11	-13	-16	-7	-22	1	-8
PM	29	26	21	30	9	-12	56
aSAT	-12	-12	-9	-18	-4	-7	-20
RETVOL	-8	-12	-7	-15	-20	-6	-27
I2A	-18	-19	-15	-23	-22	6	-38
BETA_d	-32	-37	-35	-41	-53	-8	-67
OA	-28	-27	-30	-24	-32	-21	-28
dCEQ	-22	-25	-26	-20	-34	7	-39
SIZE	-44	-44	-41	-35	-30	11	-51
SG	-17	-19	-14	-19	-20	10	-39
R_2_1	6	6	3	8	4	1	12
dSO	-34	-34	-31	-36	-31	-25	-41
S2C	-22	-21	-17	-21	-16	16	-47
SPREAD	-14	-18	-14	-18	-28	3	-35
dGS	10	11	7	11	8	6	14
DP	-25	-33	-29	-25	-51	11	-52
SUV	-6	-6	-6	-3	-5	1	-6
IPM	29	28	25	22	13	1	34
dPIA	-22	-24	-23	-21	-27	-5	-29
IDIOV	7	4	5	3	-7	-7	2
BEME	-48	-52	-48	-47	-54	-6	-70
R_36_13	-44	-46	-43	-42	-46	-20	-56

	full	paper	1964–87	1988–2011	1964–79	1980–95	1996–2011
aSIZE	-22	-22	-19	-20	-13	6	-28
Q	-45	-49	-44	-44	-52	-2	-66

The value ratios reverse between halves, the profitability spreads shrink, and momentum grows; book-to-market, Q, size, long-term reversal and the investment-type sorts are the stable core. The 1980–1995 column is flat for almost everything.

What it does to the firm-level wedges

The direct mapping refitted on each subsample’s portfolio wedges, applied to the same firm-months, against the full-sample firm wedges (sd 28 pp).

Fitted on	corr	rank	mean diff	sd diff	diff > 10	corr 2020–25	sd diff 2020–25
full 1964–2011	1.00	1.00	+0.0	0.0	0%	1.00	0.0
paper 1964–2002	1.00	1.00	+0.5	2.4	0%	1.00	2.0
first half 1964–1987	0.96	0.96	+4.8	7.5	27%	0.96	7.0
second half 1988–2011	0.92	0.91	+1.3	11.9	40%	0.91	10.9
third 1964–1979	0.91	0.90	+12.1	12.1	63%	0.90	11.0
third 1980–1995	0.44	0.42	+30.9	25.2	85%	0.28	24.1
third 1996–2011	0.90	0.89	-9.9	20.5	67%	0.90	19.3
from 1975	0.95	0.94	+5.1	9.2	35%	0.94	8.6
from 1980	0.91	0.90	+8.1	11.9	50%	0.89	11.0
from 1985	0.91	0.91	+4.6	11.5	42%	0.91	10.6
from 1990	0.92	0.91	-0.9	12.9	43%	0.91	11.9
from 1995	0.90	0.90	-7.4	18.8	63%	0.90	17.6
from 2000	0.75	0.74	+1.7	21.2	63%	0.75	18.6

The ranking of firms is robust to the sample: correlations 0.90 to 0.96 for every window except the 1980–1995 third and the post-2000 window. The magnitudes are not: refitting on the second half moves two fifths of firms by more than 10 pp, a later window more. The halves against each other at the firm level correlate 0.79 with a standard deviation of the difference of 18 pp.

Which sample

The full available sample is the headline choice. It is the only one with about three independent fifteen-year windows; every later window rests on one or two cycles, which is why the price of risk swings from 5.6 to 2.5 to 2.8 across the thirds, and the windows ending after 2008 carry a different scale (slopes 1.2 to 1.35 on the extremes). Extending the paper’s window to 2011 cohorts and 2025 cash flows is safe. Switching to a later window is not a better estimate of the same quantity: it changes the answer in the direction of the post-1990 weakness of the value ratios and the 2000 and 2008 episodes, which is regime. What the drift does argue for is showing it: the full-sample firm wedge with the two half-sample estimates as a range, or “since 1988” as a labelled alternative specification, with the statement that rankings agree at 0.9 and magnitudes differ by about ± 12 pp.

Recommendation

Use the penalised direct regression of portfolio wedges on portfolio characteristic ranks, fitted on verified characteristics only, with all ten deciles. Report firm-level wedges together with an indication of how far a firm's characteristics sit outside the range the portfolios span, because that is where the magnitudes stop being trustworthy and no portfolio-level criterion can detect it.